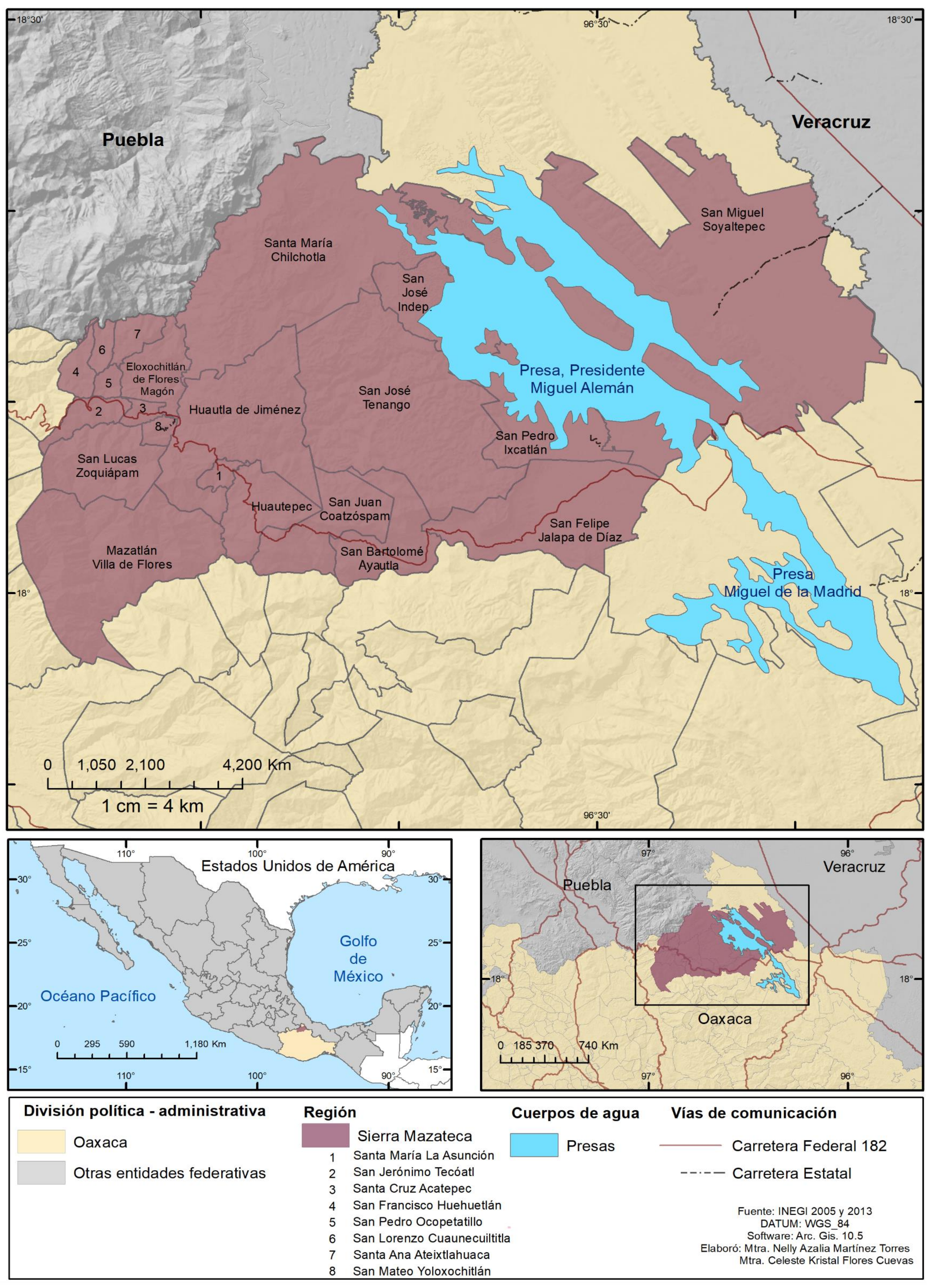
“MAZATEC WATER MANAGEMENT AND CULTURE: Loma Chapultepec, Huautla de Jiménez, Oaxaca, México”

Autora: Mtra. Celeste Kristal Flores Cuevas. Asesor: Dr. Egidio Francisco Tentori.



Introduction:

Water management is challenging as it is location-specific, depending on the uniqueness of each place, therefore there is no single mechanism for the design of a unified water management model. Likewise, water management cannot be limited to a collection of mainly technical-administrative components (planning, development, management, and distribution of water), as water management goes beyond categories and material values which see water only as a resource. Water for Indigenous peoples has a deep meaning and value associated with their worldviews and perceptions of the world and nature (Racancoj, 2011; Mazabel and Mendoza, 2012). Therefore, each social group has specific conceptions of what water is, rooted in their historical practices and their relationships with water.

Mazatec communities are characterized by managing their water in an institutionalized (technical-administrative) and communitarian (with cultural aspects) manner. However, due to national projects in the mid-twentieth century, the strategies and self-management practices of the Mazatec people have deteriorated so that both the water committees and other community positions have lost their validity within the local political system (Garcia, 2012, p.30). Therefore, this work examines aspects of Mazatec knowledge related to water to highlight the importance of weaving biocultural aspects to enable proper water management, with the understanding that there is no unifying management model, but instead thinking in terms of a plurality of efforts, where biocultural aspects are integrated, and validated to new water needs to achieve the demands of the Sustainable Development Goals (SDGs) for the year 2030.

Methodology:

Qualitative research was conducted in Loma Chapultepec, a Mazatec community located in the municipality of Huautla de Jimenez, within the Sierra Mazateca (see map). The analysis included a biocultural- and ethnoterritory-based approach with the aim of understanding the link between culture and the environment. Field research was conducted as part of a Master’s thesis from 2016 to 2019, and later in 2022 and 2023, where field tours were conducted using the ethnographic method, including interviews and participatory mapping workshops with children, youth, and adults.

Where do most indigenous communities live?

Worldwide

- **Estimated 476 million people (6% of the world's population)**
- **They occupy a quarter of the planet's surface by territory**
- **Safeguarding 80% of the world's biodiversity**

In Mexico

- **25.7 million people self-identify as indigenous**
- **4th most biologically diverse nation in the world and most of it located in indigenous territories**
- **Indigenous territories capture 23.3% of the country's water and contain the main hydroelectric dams**

Source: World Bank, 2023; National Indigenous Peoples Program 2018-2024.

What are the central axes for Mazatec water management?

1. Organization and Operation

The social organization of water in the Loma Chapultepec community is governed by four axes, which are indispensable for the functioning in the management and operation of water systems.

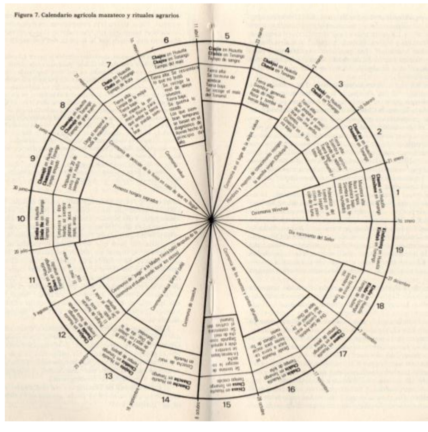
- The community assembly: serves as the guiding axis for the community, attended by adults, youths, and even children. Through this assembly decisions are made about the needs of the town, including water-related matters.
- Water Committees: the assignment of positions is done through voting in the assembly. Responsible people, often characterized by being descendants of families who in the past have helped realize community projects, are desirable candidates. They serve as the mediators between the municipal government and the community.
- Water distribution manager: represents the only remunerated role and is appointed by the community assembly and the users. Their permanence in this role depends on the users and the local government.
- Faena or *xabasén:* consists of giving time and labor for the benefit of the whole community, it is an important social encounter that marks a social recognition among families and mainly among men.

2. Biocultural Aspects

Within Mazatec water management practices, for the Mazatec people the relationship with space and territory, the transmission of knowledge through oral histories, and ceremonies or rituals are important. Mazatec water management practices include:

- Territory: the Mazatec people acknowledge that territory is shared. Springs, streams, and rivers are understood as delicate places and not respecting these water bodies, can make humans suffer spiritual illnesses, which in turn manifests in physical form.
- The inseparable relationship between the sacred mountain Chikon Tokoxo and water, from which the notion of territory in the entire Sierra Mazateca is derived. Mountains represent water reserves that in turn serve as the dwelling places of the deities and guardians of the territory and its inhabitants.
- The transmission of oral histories: the Mazatec cosmovision is centered on water, through stories the Mazatec world and the territory they inhabit are recognized.
- The main sacred ceremonies, such as weddings and baptisms, are consecrated through water.

3. Mazatec Agricultural Calendar



The Mazatec agricultural calendar expresses the wealth of local knowledge and oral wisdom that has been passed down through generations.

- Consists of 18 twenty-day periods and 5 days for rest.
- Provides specific times for different activities.
- It gives meaning to the Mazatec deities, most of which are related to water.
- More than a third of this calendar focuses on water, including water-related festivals, rituals, ceremonies, and pilgrimages.

Conclusions:

Indigenous communities confront multiple challenges, one of them being water management. Grandiose plans for hydraulic megaprojects have left more consequences than solutions in community economies, social structures, and above all, their thinking and worldview. Indigenous communities have realized that so far technocratic solutions have not solved our internal needs by failing to account our biocultural principles and values. Although there has been an increasing validation of indigenous peoples at an international level, government programs and institutions as well as academic initiatives have fallen short. Therefore, it is necessary to put ancestral knowledge into practice again and interweave this knowledge with new technologies, with the understanding that community needs are not neglected.

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Nindó Tokoxo, most important mountain for the Mazatecs.